Gauge Theory Seminar: The Classical Bauer-Furuta Invariant

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1 Seiberg-Witten-Theory

We start with a 4-manifold with a -structure $\mathfrak{s}$. Recall that this is equivalent to the data of a Hermitian bundle S together with a Clifford multiplication $\rho : T^*M \to \mathrm{End}_{\mathbb{C}}(S)$. It turns out S splits into two Hermitian subbundles S^+ and S^-, on which $\rho(a) : S^{\pm} \to S^{\mp}$ induces isomorphism for any $a \neq 0 \in T^*X$.

In the classical Seiberg-Witten setup [Wit94, Mor96], we fix (or not fix) a reference -connection A and write the Seiberg-Witten map as

$$\begin{aligned} \mu : (A + i\Omega^1(X;\mathbb{R})) \oplus \Gamma(S^+) &\to \Gamma(S^-) \oplus \Omega^2_+(X;\mathbb{R}) \oplus \Omega^1(X;\mathbb{R})/\mathbb{R} \\ (A + ia, \phi) &\mapsto (d^+a - q(\phi) + \frac{1}{2}F^+_{A^t}, D^+_A\phi + \rho(a)\phi, d^*a) \end{aligned}$$

where d^+a is the self-dual part of da, $q(\phi)$ is the ρ^{-1} trace free part of $\phi\phi^* \in \Gamma(\mathrm{End}_{\mathbb{C}}(S))$, $F^+_{A^t}$ is the self-dual part of F_{A^t}, where A^t is the induced connection on $\det(()S^+)$, D^+_A is the Dirac operator on positive spinors, and d^* is the codifferential.

The index of the Dirac operator is computed by the Atiyah–Singer index theorem [AS68]. Write $L = \det(()S^+)$, so $c_1(L) = c_1(\mathfrak{s})$, and let $\sigma(X)$ be the signature.

Theorem 1.1 ([AS68]). *The complex index of $D^+_A : \Gamma(S^+) \to \Gamma(S^-)$ is*

$$\mathrm{ind}_{\mathbb{C}}\, D^+_A = \frac{c_1(L)^2 - \sigma(X)}{8},$$

where $c_1(L)^2$ means $\langle c_1(L)^2, [X]\rangle$. In particular $\mathrm{ind}_{\mathbb{R}}\, D^+_A = \frac{c_1(L)^2-\sigma(X)}{4}$, independent of A.

Sketch. Locally $S^+ = S^+_0 \otimes L^{1/2}$ for a genuine spinor bundle S^+_0, so the index theorem gives

$$\mathrm{ind}_{\mathbb{C}}\, D^+_A = \int_X \hat{A}(X)\, e^{c_1(L)/2} = \int_X \Big(\frac{c_1(L)^2}{8} - \frac{p_1(X)}{24}\Big),$$

and $p_1(X)[X] = 3\sigma(X)$ by the Hirzebruch signature theorem. Integrality is consistent with $c_1(L)$ being characteristic: $c_1(L)^2 \equiv \sigma(X) \pmod 8$. □

In the Bauer-Furuta paper [BF04], the equation they use is a globalized version of Seiberg-Witten map, where the base is the space of connections as follows:

$$\begin{aligned} \mathcal{A} &:= (A + \ker(d)) \times (\Omega^1(X;\mathbb{R}) \oplus \Gamma(S^+))/\mathcal{G}_0 \\ \mathcal{C} &:= (A + \ker(d)) \times (\Gamma(S^-) \oplus \Omega^2_+(X;\mathbb{R}) \oplus H^1(X;\mathbb{R}) \oplus \Omega^0(X;\mathbb{R})/\mathbb{R})/\mathcal{G}_0 \end{aligned}$$

and the Seiberg-Witten map is

$$\begin{aligned} \mu : \mathcal{A} &\to \mathcal{C} \\ (A, \phi, a) &\mapsto (A, D_{A+a}\phi, F^+_{A+a} - \sigma(\phi), a_{harm}, d^*a) \end{aligned}$$

Claim 1.2. *$\mathcal{A}$ and $\mathcal{C}$ are S^1-equivariant Hilbert bundles over $Pic(X)$ and the Seiberg-Witten map μ is an S^1-equivariant, fiber-preserving map.*

Now $\mu = l + c$, where $l = D_A \oplus d^+ \oplus pr_{harm} \oplus d^*$ and $c : (\phi, a) \mapsto (0, F^+_A, 0, 0) + (a \cdot \phi, -\sigma(\phi), 0, 0)$.

Claim 1.3. *c is compact.*

Claim 1.4. *The linear part l is Fredholm at any $A \in Pic(X)$, so $\mu = l + c$ is a compact perturbation of a family of linear Fredholm operators, and*

$$\operatorname{ind} l = \operatorname{ind} D - \underline{H^2_+(X; \mathbb{R})}$$

where the underline means the trivial bundle over $Pic(X)$.

Proof. 1. $\operatorname{ind} D_A$ has its own domain and codomain, so that's fine.
2. $d^+ \oplus d^*$ has kernel $\mathcal{H}^1(X)$, the harmonic group, and cokernel $\mathcal{H}^2_+(X)$.
3. The addition of this harmonic projection kills the kernel $\mathcal{H}^1(X)$ in the domain. □

Theorem 1.5 ([BF04, Thm. 3.1]). *μ satisfies the boundedness condition. In particular, it defines a stable cohomotopy class $[\mu] \in \pi^{b^+_2}_{S^1}(Pic(X); \operatorname{ind} D) = \mathbb{S}^{b^+_2(X)}_{S^1, H}((\operatorname{ind} D_X))$.*

The Bauer Furuta recovers the integer Seiberg-Witten invariant in the following way:

Theorem 1.6 ([BF04, Thm. 3.3]). *If $b^+_2 > b_1 + 1$, the choice of homology orientation $H^1(X; \mathbb{R}) \oplus H^2_+(X; \mathbb{R})$ determines a map*

$$t : \pi^{b^+_2}_{S^1}(Pic(X); \operatorname{ind} D) \to \mathbb{Z}$$

Proof. A stable map can be represented by a pointed equivariant map

$$\mu : TF \to V^+$$

where $\operatorname{ind} D = TF - Pic(X) \times V'$ and $V = (V' \oplus H^2_+(X; \mathbb{R}))^+$.

Then we perturb such that TF^{S^1} are sent to a subspace of $(V^+)^{S^1}$ of codimension $b^+_2 - b_1 \geqslant 2$. Then we can further perturb it such that a generic point in the complement of $(V^+)^{S^1}$ is a manifold M with free S^1-action on it. The homology orientation induces orientation on F and V, then we have a relative orientation between TF and V, which will induce an orientation on M. Note that

$$\dim M = \operatorname{ind}_{\mathbb{R}} D + b_1 - b^+_2$$

why b_1? Because we are taking a family with base $Pic(X)$, which has dimension b_1. Plugging in the index of D,

$$\dim M = \frac{c_1(L)^2 - \sigma(X)}{4} + b_1 - b^+_2, \qquad \dim M/S^1 = \frac{c_1(L)^2 - 2\chi(X) - 3\sigma(X)}{4},$$

which is the usual expected dimension of the Seiberg–Witten moduli space (use $\chi = 2 - 2b_1 + b_2^+ + b_2^-$ and $\sigma = b_2^+ - b_2^-$).

Now take $L = M \times_{S^1} \mathbb{C}$, this line bundle has first chern class in $H^2(M/S^1)$, so if $\dim M = 2k+1$, we can just take

$$c_1(L)^k[M/S^1]$$

and this is an integer. □

2 Permutation of Manifolds

Theorem 2.1 ([Bau04, Thm. 1.1]). $[\mu_{X\#Y}] = [\mu_X] \wedge [\mu_Y]$.

Sanity Check: The wedge sum on the right is a map

$$\begin{aligned}\wedge : [\mathbb{S}^{\operatorname{ind} D_X}, \mathbb{S}^{b_2^+(X)}] \times [\mathbb{S}^{\operatorname{ind} D_Y}, \mathbb{S}^{b_2^+(Y)}] &\to [\mathbb{S}^{\operatorname{ind} D_X} \wedge \mathbb{S}^{\operatorname{ind} D_Y}, \mathbb{S}^{b_2^+(X)+b_2^+(Y)}] \\ &\cong [\mathbb{S}^{\operatorname{ind} D_X \boxtimes \operatorname{ind} D_Y}, \mathbb{S}^{b_2^+(X)+b_2^+(Y)}] \\ &= [\mathbb{S}^{\operatorname{ind} D_{X\#Y}}, \mathbb{S}^{b_2^+(X\#Y)}]\end{aligned}$$

So implicitly, the authors showed that $\operatorname{ind} D_X \boxtimes \operatorname{ind} D_Y \cong \operatorname{ind} D_{X\#Y}$.

Construction 2.2 ([Bau04]). Given n connected 4-manifolds, $\bigsqcup X_i$ is also a 4-manifold. Choose a separating neck $N_i \cong [-L, L] \times S^3$ in each connected component X_i. Then X_i splits into two parts, we call one the positive part and the negative part. We may permute the positive parts of X_i's by some permutation τ. This also induces an identification of $Pic(X) \to Pic(X^\tau)$ and equivariant bundle isomorphisms

$$\mathcal{A} \to \mathcal{A}^\tau$$
$$\mathcal{C} \to \mathcal{C}^\tau$$

More specifically, for any vector bundle $E \to X$, we fix an identification $E|_{\sqcup N_i} \cong \oplus_{i=1}^n F$. Then under the permutation, the vector bundles glued up along S^3's to give E^τ. We would like this permutation to give a map $V : \Gamma(E) \to \Gamma(E^\tau)$.

The problem is a smooth section might not be glued up smoothly under the permutation. So we have to smooth it. Here's the way the authors did: given a section e, we can write $e = (e_1, ..., e_n)$, where e_i is the section on X_i. Then choose a bump function ψ that vanishes on $[-L, -1]$ and 1 on $[1, L]$, then choose a path $\gamma : [0, 1] \to SO(n)$ that starts at the identity and ends at τ. Here we need $\tau \in A_n$! Then we define

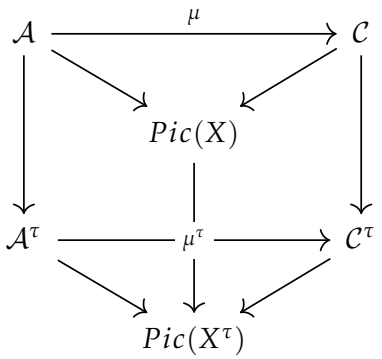
$$e^\tau = (\gamma \circ \psi)(e)$$

and we get a map $V : \Gamma(E) \to \Gamma(E^\tau)$.

Claim 2.3. *The above map is an isomorphism of Hilbert spaces.*

The above maps are all constructed in this way.

Theorem 2.4 ([Bau04]). *The following diagram commutes up to a homotopy through Fredholm maps.*



Therefore, the induced cohomotopy map $[\mu]$ and $[\mu^\tau]$ are the same. So we have

$$\bigwedge[\mu_{X_i}] = [\mu_X] = [\mu_{X^\tau}] = \bigwedge[\mu^\tau_{X_i}]$$

Now we can form the disjoint union of $X \sqcup Y$, then by doing a transposition, we get $X\#Y \sqcup S^4$. The previous theorem tells us that

$$[\mu_X] \wedge [\mu_Y] = [\mu_{X\#Y}] \wedge [\mu_{S^4}]$$

Theorem 2.5. $[\mu_{S^4}] = [\mathrm{id}]$.

So we have the promised theorem.

References

[AS68] M. F. Atiyah and I. M. Singer, *The index of elliptic operators: III*, Ann. of Math. (2) **87** (1968), 546–604.

[BF04] S. Bauer and M. Furuta, *A stable cohomotopy refinement of Seiberg–Witten invariants: I*, Invent. Math. **155** (2004), no. 1, 1–19.

[Bau04] S. Bauer, *A stable cohomotopy refinement of Seiberg–Witten invariants: II*, Invent. Math. **155** (2004), no. 1, 21–40.

[Fur01] M. Furuta, *Monopole equation and the $\frac{11}{8}$-conjecture*, Math. Res. Lett. **8** (2001), no. 3, 279–291.

[Mor96] J. W. Morgan, *The Seiberg–Witten Equations and Applications to the Topology of Smooth Four-Manifolds*, Mathematical Notes **44**, Princeton University Press, 1996.

[Wit94] E. Witten, *Monopoles and four-manifolds*, Math. Res. Lett. **1** (1994), no. 6, 769–796.